

**Time:** 5 minutes**Outcome:** Distinguish nominal and real GDP and calculate the GDP deflator.**Difficulty:**

Intermediate

Real versus Nominal GDP



THE CORE IDEA

Nominal GDP values current production at current prices, while real GDP values it at base-year prices. Multiply current quantities by the designated base-year prices and add across final goods. GDP is a strong measure of market production for its purpose, but it omits some dimensions of welfare. Compare nominal and real GDP to separate price changes from changes in production. Real-GDP growth equals the change in real GDP divided by the initial real GDP.



HOW TO RECOGNIZE IT

- Nominal GDP uses current-year prices and current-year quantities.
- Real GDP uses base-year prices with current-year quantities.
- A change in nominal GDP can reflect prices, quantities, or both; a change in real GDP reflects quantities.
- Use the GDP deflator to compare nominal GDP with real GDP for the same year.



WATCH OUT

Do not use the new level rather than the initial real-GDP level as the percentage-growth denominator. Real-GDP growth equals the change in real GDP divided by the initial real GDP.



WORKED EXAMPLE: SEPARATING PRICE AND OUTPUT CHANGES

One-good economy: price and output

	Quantity	Price	Nominal GDP	Real GDP
Base year	100	\$5	\$500	\$500
Current year	110	\$6	\$660	\$550

An economy produces 100 units in the base year at \$5 each and 110 units this year at \$6 each. Nominal GDP this year is $110 \times \$6 = \660 . Real GDP at base-year prices is $110 \times \$5 = \550 . The GDP deflator is $(\$660 / \$550) \times 100 = 120$, separating the price increase from the output increase.



CHECK YOURSELF

Is real GDP calculated by simply subtracting the inflation rate from nominal GDP?

**READY?**

Return to the game and master this concept.